Remote-Sensing Image Scene Classification With Deep Neural Networks in JPEG 2000 Compressed Domain

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***Abstract*— To reduce the storage requirements, remote-sensing (RS) images are usually stored in compressed format. Existing scene classification approaches using deep neural networks (DNNs) require to fully decompress the images, which is a computationally demanding task in operational applications. To address this issue, in this article, we propose a novel approach to achieve scene classification in Joint Photographic Experts Group (JPEG) 2000 compressed RS images. The proposed approach consists of two main steps: 1) approximation of the finer resolution subbands of reversible biorthogonal wavelet filters used in JPEG 2000 and 2) characterization of the high-level semantic content of approximated wavelet subbands and scene classification based on the learned descriptors. This is achieved by taking codestreams associated with the coarsest resolution wavelet subband as input to approximate finer resolution subbands using a number of transposed convolutional layers. Then, a series of convolutional layers models the high-level semantic content of the approximated wavelet subband. Thus, the proposed approach models the multiresolution paradigm given in the JPEG 2000 compression algorithm in an end-to-end trainable unified neural network. In the classification stage, the proposed approach takes only the coarsest resolution wavelet subbands as input, thereby reducing the time required to apply decoding. Experimental results performed on two benchmark aerial image archives demonstrate that the proposed approach significantly reduces the computational time with similar classification accuracies when compared with traditional RS scene classification approaches (which requires full image decompression).**

***Index Terms*— Compressed image domain, deep neural networks (DNNs), joint photographic experts group (JPEG) 2000, remote sensing (RS), scene classification.**

I. Introduction

ADVANCEMENTS in satellite technologies have led to a huge increase in the volume of remote-sensing (RS) image archives up to an unprecedented level. Subsequently, the extent of information that can be inferred from these massive archives is also increasing, enabling applications. Considering the volume of such massive archives as well as the complexity of RS images, developing efficient scene classification methods is one of the most important research topics in RS.

Scene classification aims to assign a class label to a given RS image based on the analysis of descriptors that characterize its semantic content. The performance of any scene classification method mainly depends on its ability to characterize a given RS image with efficient feature representations. In the last decade, several handcrafted features were introduced in the RS literature [1]–[8]. However, exploiting the aforementioned handcrafted features for scene classification tasks in massive RS image archives that contain petabytes of data is inefficient, time-demanding, and computationally complex. Moreover, the discriminative power of these handcrafted features is often shallow and requires human intelligence, which incurs additional labor costs. To address these limitations, in recent years, several deep learning (DL) algorithms were introduced in RS community [9]–[13]. Among them convolutional neural networks (CNNs) have demonstrated their remarkable ability to learn the high-level semantic content of RS image resulting in high classification accuracies. These networks hierarchically learn the intrinsic patterns within the images through several convolution and pooling operations to obtain distinctive feature descriptors. Several studies have shown very good performance of CNNs in the RS domain for several applications, including scene classification [4], [9], [14]. In the early years, CNN models were trained from scratch with a considerable amount of training data. However, recent experiments have shown that using pretrained DL networks (e.g., ImageNet [15], VGG16 [16], GoogLeNet [17], CaffeNet [18]) in RS domain has remarkably improved classification performance. In [19], it is shown that the classification accuracy of a CNN model with a considered pretrained network outperforms what obtained when a model is trained from scratch. This mainly depends on the small number of training samples usually available in RS applications.

Recently, several studies were conducted to enrich the discriminative power of the image descriptors obtained through conventional CNN models. As an example, combining features

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obtained at multiple resolutions to represent a scene provides significant improvement in classification accuracies [20]. Although this approach is rotation as well as translation invariant, it is time-demanding to train images at several resolutions using a simple CNN model. Zheng *et al.* [21] proposed a deep representation where the multiscale features are obtained from the image feature maps using multiscale pooling (MSP) to improve the classification performance at a faster rate. To improve the classification performance, features obtained from relevant image areas can be considered. This can be done by introducing the attention mechanisms. In [22], local image descriptors are obtained with recurrent attention to perform multilabel RS scene classification. Attention mechanism avoids irrelevant areas and focus to obtain features from relevant image regions thereby reducing the number of parameters as well as the computational time required to perform a specific task. Guo *et al.* [23] have proposed a network where both local and global features are learned through a local-global attention framework.

To mitigate the problem of overfitting that generally arises due to the limited availability of annotated data, generative adversarial networks (GANs) became popular to address scene classification problems in RS. GANs learn the hidden structure in the given input data and consists of a generator (that learns the semantic contents of the data) and a discriminator (that classifies the generated and input images) [24]–[29]. MARTA-GAN proposed in [27], was one among the first efforts to learn feature representations to perform unsupervised aerial scene classification. The performance of GAN models depends on the quality of the modeling of the structure of generated images. In [28], GAN-NL has proposed to effectively model the nonlocal dependencies in the generated images and has shown remarkable improvements in the classification accuracies. Roy *et al.* [26] proposed a semantic fusion GAN, where the feature representations are obtained using a standard deep convolutional GAN (DCGAN) combined with an external deep network. In [29], two CNNs are integrated that serve as the generator and discriminator model of GAN to classify hyperspectral images. The authors proposed 1-D-GAN and 3-D-GAN that are used to classify the spectral and spatial-spectral information, respectively. Although the current state-of-the-art GAN networks demonstrate an improvement in the classification performance, their optimization and training are time demanding.

To reduce the storage required for huge amounts of data, RS images are compressed before storing them in the archives [30]. Several image compression algorithms such as differential pulse code modulation (DPCM), adaptive DPCM (ADPCM), Joint Photographic Experts Group (JPEG), and JPEG 2000 were introduced in RS [30]. Among several compression algorithms, JPEG 2000 [31] became very popular in RS due to its multiresolution paradigm, scalability, and high compression ratio. JPEG 2000 algorithm is used to compress RS images acquired by most of the recent applied satellites (such as Sentinel-2 and PRISMA) in their archives. Thus, before performing any scene classification of images in compressed archives, the image decompression task should be performed. This is computationally demanding and ineffective when considering real large-scale RS image archives that may contain petabyte scale data. Considering the complexity and size of RS image, the amount of time required to decompress any image cannot be neglected in real large-scale RS image archives.

To address the aforementioned limitation, in this article, we propose a novel approach based on deep neural networks (DNNs) for scene classification of compressed RS images. We assume that the images in the archive are compressed using JPEG 2000. The proposed approach aims to minimize the amount of decompression required for the classification of compressed RS images while maintaining similar accuracies when compared to the conventional state-of-the-art scene classification approaches. To achieve this, the proposed approach consists of two main steps: 1) approximation of the finer (higher) resolution wavelet subbands used in JPEG 2000 compression algorithm and 2) feature extraction and scene classification of the approximated finer resolution wavelet subbands. The proposed approach initially obtains the codestreams associated with the coarsest resolution wavelet subbands of the considered JPEG 2000 compressed RS image. Then, in the first step, finer resolution wavelet subbands (image) are approximated through a series of transposed convolutional layers. The second step aims to obtain the features associated with the approximated finer resolution wavelet subbands and perform scene classification based on the learned descriptors. This is obtained by introducing a loss function that learns the parameters associated with both approximation and scene classification in an end-to-end unified neural network. In the classification phase, the proposed approach requires only the codestreams associated with the coarsest wavelet resolution subbands and thereby reduces the time required to preform decompression of the images. The proposed approach explores the hierarchical multiresolution feature space in a unified framework and achieves optimal resource utilization for scene classification. The effectiveness of the proposed approach was evaluated using two different aerial benchmark image archives: NWPU-RESISC45 [32] and AID [19]. It is noted that the aim of this study is not to introduce a compression algorithm but to propose a novel DL approach that requires minimally decoded wavelet subband information to obtain optimal classification, thereby significantly minimizing the computational cost of the decoding step.

The remainder of this article is organized into five sections: Section II gives a brief review of the related works. Section III explains the proposed approach. Section IV describes the data sets and the experimental setup, while Section V illustrates the experimental results with discussion. Finally, Section VI draws the conclusion of the work.

II. Related Works

A. JPEG 2000 Algorithm

JPEG 2000 is one among the most popular compression algorithms and image coding systems that rely on the wavelet-based approach. Fig. 1 illustrates the block scheme

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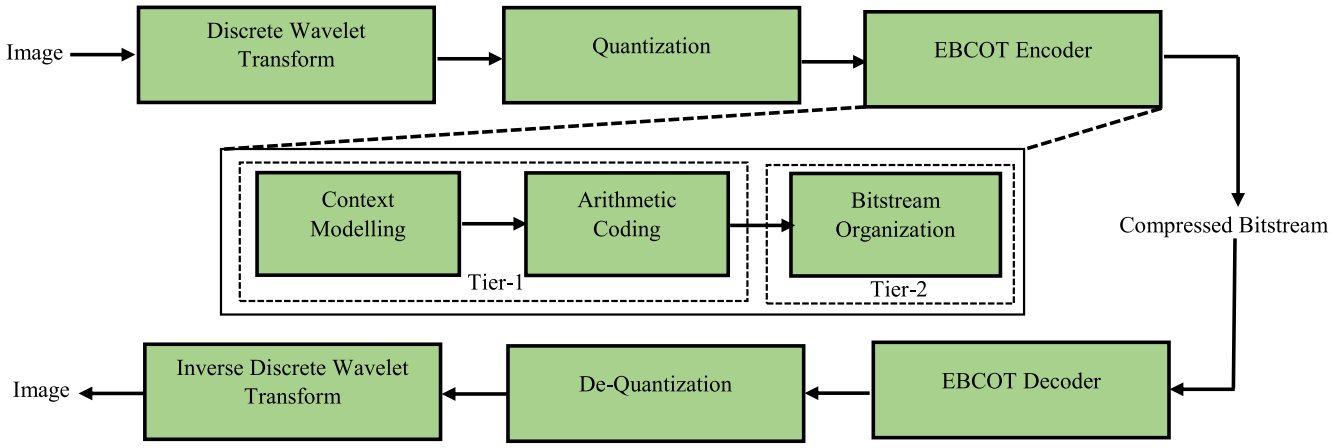


Fig. 1. General block scheme of the JPEG 2000 compression and decompression algorithm.

in the encoding and decoding of JPEG 2000 compression algorithm. It includes: 1) discrete wavelet transform (DWT); 2) quantization; and 3) embedded block coding with optimized truncation. DWT when applied to a given image helps to achieve multiresolution image description and decomposes the given image into one approximation (LL) subband and three detail subbands (LH, HL, and HH). The LL subband is further decomposed to approximation and detail subbands, if one requires more than one decomposition level. The wavelet coefficients of each subband are quantized using a step-size quantizer as selected by the user. As we consider lossless compression for the images in the archive, we do not perform quantization and thus this step is neglected. The subbands are then subdivided into rectangular blocks referred as precinct and each precinct is again subdivided into codeblocks that are usually of size 64×64. The minimum size of a codeblock should be 32×32. These codeblocks, which are represented using bitplanes, are then entropy coded using the EBCOT algorithm. Entropy coding in JPEG 2000 compression algorithm is separated into Tier-1 and Tier-2 coding. In Tier-1 coding, each codeblock is coded using: 1) context modeling and 2) arithmetic coding. The probability of occurrence of a particular bit can be determined from the contextual information, which is obtained from three passes: significance propagation pass, refinement pass, and cleanup pass. In Tier-2 coding, the bitstreams obtained after arithmetic coding are organized into several packets and layers. A given packet includes the codestream associated with a particular precinct and allows to access data based on the quality, resolution, or band. This packet structure organization allows to access progressively and hierarchically to the information of a given image.

B. JPEG 2000-Based Feature Descriptors

Features that can be obtained from JPEG 2000 compressed images for scene classification are broadly divided into two categories: header-based and wavelet-based features. Header-based features are directly obtained from the bitstreams of a JPEG 2000 compressed image, where one can acquire information, such as the number of bytes B used to entropically encode a given image or the maximum number of significant bitplanes MB in a given codeblock [33]. Mallat and Hwang [34] illustrates that these features represent the singularities present in an image, which can be utilized for image classification/retrieval.

Wavelet-based features are obtained after the partial decompression of a given compressed image. They efficiently model spectral, texture, and shape information obtained from the approximation and detail (horizontal, vertical, and diagonal) wavelet subbands. The problem of obtaining features from wavelets has been studied extensively during the last decade. By observing that the detail subbands have a near-Gaussian behavior, most of the early research works on image classification/retrieval focused on modeling the detail wavelet subbands using generalized Gaussian distribution (GGD) [35], Gaussian mixture model (GMM) [36], generalized gamma density (GTD) [37], and their variants. Although these statistical representations are highly discriminative, they are computationally complex and time-consuming. The energy and mean descriptor obtained from the detail subbands were also used as the texture features [38]. In addition, features extracted by the co-occurrence matrix (i.e., contrast, homogeneity, energy, variance, and correlation) were considered to model the texture features. The histogram obtained from the probability of joint distribution of bitplanes that are obtained directly from the JPEG 2000 codestreams have been used for image retrieval [38]. In [39], the spectral information obtained from the approximation subband exploited to perform image retrieval in a JPEG 2000 compressed archive. In [40], the moduli as well as the angle of the wavelet coefficients obtained from the horizontal and vertical wavelet subbands were found very effective in modeling the edge features for image classification. In [41], morphological operations such as dilation and erosion applied to wavelet subbands were used to obtain shape-based features to perform image retrieval from JPEG 2000 compressed images. However, these handcrafted features are unable to capture the high-level semantic information when compared with deep features. Thus, this work focuses on developing a novel approach that benefits from DNNs to achieve efficient scene classification performance in JPEG 2000 compressed image archives.

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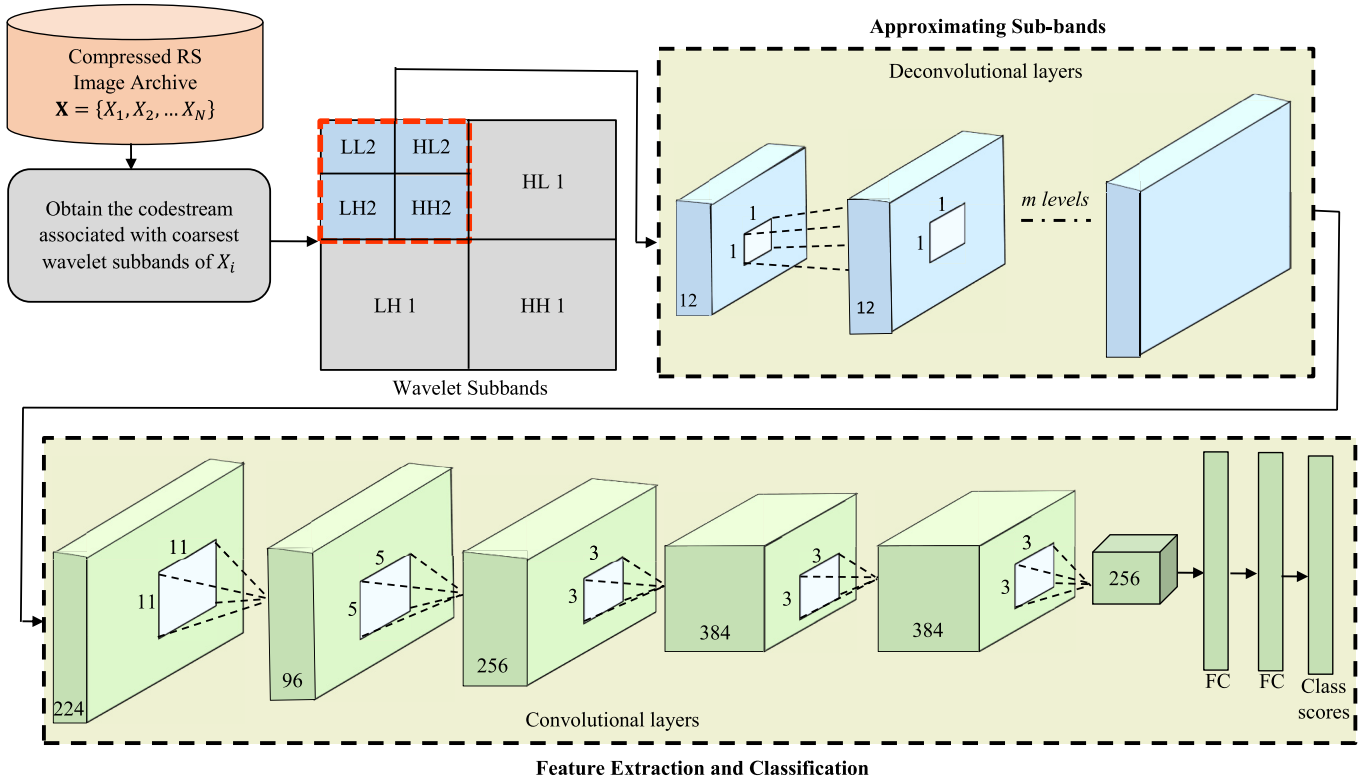


Fig. 2. Block scheme of the proposed approximation approach in the compressed domain.

III. Proposed Scene Classification Approach in the JPEG 2000 Compressed Domain

A. Problem Formulation

Let $\mathbf{X} = \{X_i\}_{i=1}^{N}$ be an archive that contains N JPEG 2000 compressed RS images, where X_i represents the ith image. The main objective of the proposed approach is to assign a class label $y_i \in \mathbf{Y}$, where $\mathbf{Y}$ is a set of class labels, to a given input image $X_i \in \mathbf{X}$. Let us assume that all the images in the archive are decomposed up to L resolutions. Each image in the archive will be represented as one approximation subband and $3L$ detail subbands (i.e., horizontal, vertical, and diagonal). In JPEG 2000 compressed image archive, the straightforward approach to perform scene classification is to: 1) apply entropy decoding to the code streams associated with all the images in the archive and 2) obtain the image descriptors. However, decoding all the images from a compressed archive is time-demanding and computationally expensive. Thus, we propose a novel approach based on DNNs that efficiently approximates a decompressed image to perform scene classification in a large-scale JPEG 2000 compressed image archive. Our objective is: 1) to implement a novel DL approach that performs scene classification in the compressed domain with minimal decompression and 2) to reduce the computational time when compared to models that require fully decoded images. To achieve this, the proposed approach consists of two main steps: 1) approximation of the finer (higher) resolution wavelet subbands used in JPEG 2000 compression algorithm and 2) feature extraction and scene classification of the approximated finer resolution wavelet subbands. Fig. 2 shows the block scheme of the proposed approach and each step is explained in Sections III-B and III-C.

B. Approximation of the Wavelet Coefficients

This step aims to approximate the finer (higher) resolution wavelet subbands (or the image itself) through a series of transposed convolutional layers. To this end, the proposed approach considers m transposed convolutional layers, where m corresponds to the number of wavelet decomposition levels that were initially used to compress a given image $X_i \in \mathbf{X}$. Given a compressed image X_i, we initially obtain the codestream associated with the coarsest level wavelet subband (see Fig. 2) that provides the global-scale information of any given image. Let $G^L = \{a_{X_i}^L, h_{X_i}^L, v_{X_i}^L, \text{ and } d_{X_i}^L\}$ denote the approximation, horizontal, vertical, and diagonal subbands of an image X_i at the Lth wavelet decomposition level (coarsest wavelet subband). Let $A^{L-1} = \{a_{X_i}^{L-1}, h_{X_i}^{L-1}, v_{X_i}^{L-1}, \text{ and } d_{X_i}^{L-1}\}$ be the finer level approximated subbands at level $L-1$.

We consider "convolution" operation as a matrix multiplication between the flattened input and a sparse matrix C. As an example, G^L can be obtained by applying convolution operation on A^{L-1} as follows:

$$G^L = C \cdot A^{L-1}. \tag{1}$$

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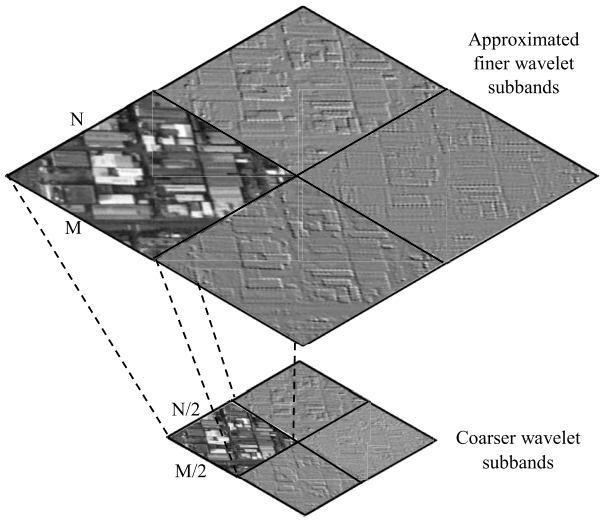


Fig. 3. Example of approximation of a finer level wavelet subbands using transposed convolution.

The nonzero elements in the sparse matrix C can be constructed using the kernel coefficients of the convolution operation as follows:

$$C = \begin{bmatrix} k_{11} & \dots & k_{1q} & 0 & \dots & k_{2q} & \dots & k_{pq} & 0 & \dots \\ 0 & k_{11} & \dots & k_{1q} & 0 & \dots & k_{2q} & \dots & k_{pq} & \dots \\ 0 & 0 & k_{11} & \dots & k_{1q} & 0 & \dots & k_{2q} & & \dots \\ & & & & & & & & & \\ \dots & & & & & & & & & \dots \\ 0 & 0 & 0 & 0 & & & \dots & & & k_{pq} \end{bmatrix} \quad (2)$$

where p and q represent the kernel size and k_{ij} is the element of the kernel (where i and j are the row and column indices of the kernel, respectively). Convolution operation takes the input matrix A^{L-1} which is then flattened into a vector and multiplies the flattened input with C. The matrix multiplication result is reshaped to obtain the final output G^L. It is worth noting that during the forward and backward passes of CNNs, convolution operations are applied with C and C^T, respectively. In computer vision, transposed convolution has proved to be an efficient algorithm that uses the gradient of the convolution operation (for a given image) to perform image restoration and reconstruction [42]. Our proposed approach approximates the finer wavelet subbands using transposed convolution as shown in Fig. 3. Given a kernel, the transposed convolution operation multiplies the flattened input vector G^L with C^T during the forward pass and multiplies with $(C^T)^T = C$ during the backward pass to obtain A^{L-1}. The finer level wavelet subbands can be obtained as

$$A^{L-1} = C^T \cdot G^L \quad (3)$$

where we swap the backward and forward passes of the convolution operation which is used in standard CNNs. Accordingly, using m transposed convolutional layers, the proposed approach allows to approximate the image A^m. For the transposed convolutional layers, if we use a stride S, padding P, and kernel size k, then the size of the approximated wavelet subbands (A^{L-1}_{size}) obtained from the coarser level wavelet subbands (A^{L}_{size}) is given by

$$A^{L-1}_{\text{size}} = S * \left(A^{L}_{\text{size}} - 1\right) + k - 2P. \quad (4)$$

The proposed approximation approach reflects the inherent multiresolution paradigm within the JPEG 2000 compression algorithm within an end-to-end unified framework. While approximating subbands, we consider two scenarios.

1) Scenario 1 (Minimal Decoding): In this scenario, the proposed approach obtains only the codestreams associated with the coarsest level (Lth level) wavelet subbands to approximate the finer level subbands (image itself). Here, the aim is to minimize the amount of decompression time required to perform scene classification by approximating wavelet subbands (image) using only the coarsest level subband. The coarsest level wavelet subband provides global-scale information of the considered image. Thus, in this scenario, although the amount of time required for decompression is significantly reduced, the quality of approximation is moderately diminished. Fig. 4 illustrates the case when the proposed approach takes the codestreams associated with 32×32 coarsest level wavelet subband to approximate the image A^m using m transposed convolutional layers.

2) Scenario 2 (Partial Decoding): In this scenario, the proposed approach takes the coarsest level (Lth level) wavelet subband information to decode the finer level ($L-1$th level) wavelet subband, which is exploited to approximate the finest level wavelet subbands (image itself). In this scenario, the amount of required decompression time is reduced moderately to achieve favorable performance, when compared to the case where the images require full decompression. The finer level wavelet subbands provide fine-scale information of a given image. Thus, the wavelet subbands (image) approximated from the finer level subbands incorporate the detailed fine-scale information that enhance the classification accuracy with moderate reduction in time. Fig. 5 illustrates the case when a 32×32 coarsest level wavelet subband is employed to decode the finer level subbands of size 64×64. Then, deconvolution is applied to the decoded finer level wavelet subbands to approximate the finest level wavelet subbands (image).

C. Feature Extraction and Classification

The feature extraction and classification step aims to obtain features from the approximated wavelet subbands (image). To this end, we consider a neural network with five convolutional layers with a number of filters similar to that of the AlexNet [43] and two fully connected (FC) layers. By modifying the feature extraction and classification steps, we can obtain powerful discriminative features. To demonstrate the effectiveness of recent DL models when used in the compressed domain wavelet subband information, we selected the ResNet50 [44] architecture to compare with the results

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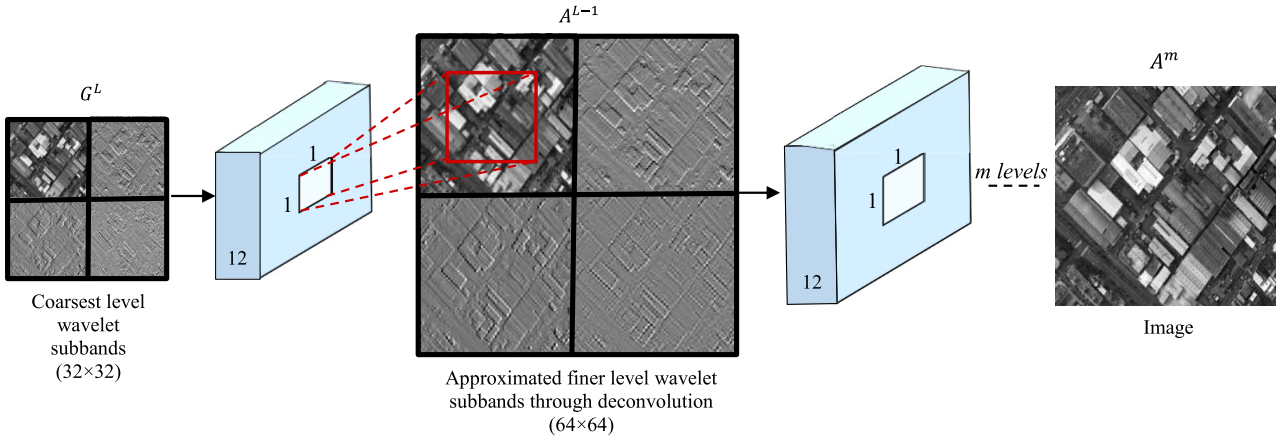


Fig. 4. Illustration of the proposed approach when the approximated image is obtained from codestream of coarsest level wavelet subband.

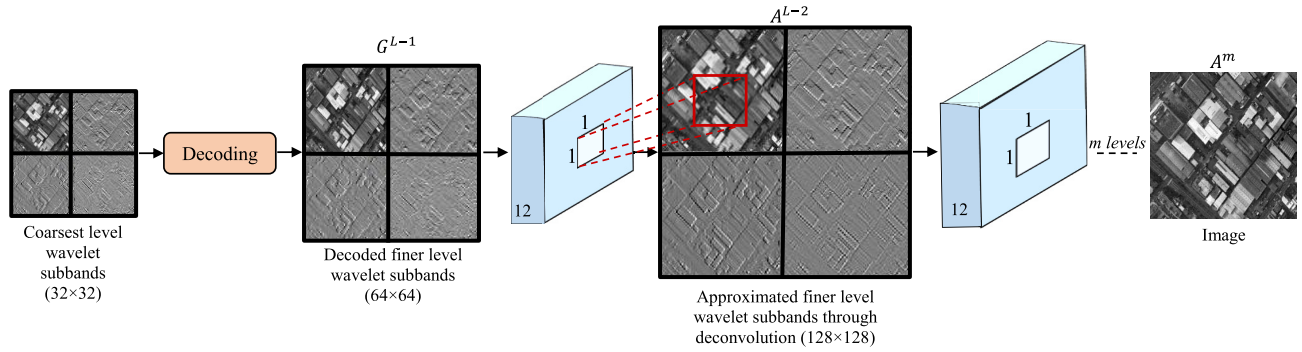


Fig. 5. Illustration of the proposed approach when the approximated image is obtained from codestream of finer level wavelet subband.

obtained by the AlexNet. Then, the output obtained from the final FC layer is mapped into classification scores. To reduce information loss, we considered zero padding and stride of 1 in each convolutional layer, which is followed by a max-pooling layer except the third and fourth.

The total loss ($\mathcal{L}_{\text{total}}$) of the proposed approach is the sum of the approximation loss ($\mathcal{L}_{\text{approximation}}$) and classification loss ($\mathcal{L}_{\text{classification}}$), which is obtained as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{classification}} + \mathcal{L}_{\text{approximation}}. \tag{5}$$

The approximation loss $\mathcal{L}_{\text{approximation}}$ is obtained by calculating the sum of mean-squared errors (mse) between the approximated wavelet subbands and decoded wavelet subbands at each level l as follows:

$$\mathcal{L}_{\text{approximation}} = \sum_{i=L}^{1} \sum_{j=1}^{M} \sum_{k=1}^{N} ||A^i(w[j,k]) - D^i(w[j,k])||^2 \tag{6}$$

where $M \times N$ represents the size of the considered wavelet subbands at level l, $w[j,k]$ denote the wavelet coefficient at position $[j,k]$ and D^i represents the decoded wavelet subband at any given level i. The classification loss $\mathcal{L}_{\text{classification}}$ is the cross-entropy loss function, which is predominantly used for scene classification problems and defined as follows:

$$\mathcal{L}_{\text{classification}} = -\sum_{i=1}^{Q} y_i \log \hat{y}_i \tag{7}$$

where $\hat{y}_i$ denote the predicted class label. To improve the classification performance, batch normalization (BN) and dropout were carried out after each convolutional layer. To overcome vanishing gradient problem, Rectified Linear Unit (ReLU) activation was used after both the convolutional and transposed convolutional layers. Section IV provides the more detailed information regarding the training details and the parameters.

IV. Data Set Description and Experimental Setup

Several experiments were performed to evaluate the performance of the proposed approach on two benchmark archives. The first one is the NWPU-RESISC45 [32] benchmark archive that consists of 31 500 images associated with 45 different categories (i.e., airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland). Each category has 700 scene classes and each image in the archive has the size of 256 × 256 with a varying spatial resolution between 0.2 and 30 m/pixel. The reader is referred to [32] for detailed information.

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TABLE I
NUMBER OF IMAGES CONSIDERED FOR EACH ARCHIVE IN THE TRAINING, VALIDATION, AND TEST DATA

Image Archive	Training	Validation	Test
NWPU-RESISC45	25200	3150	3150
AID	8000	1000	1000

The second archive is the AID [19] benchmark archive that contains 10 000 images associated with 30 different categories (i.e., airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct). Each image has the size of 600×600 pixels with a spatial resolution in the range from 0.5 to 8 m. For more detailed information, the reader is referred to [19].

To assess the effectiveness of the proposed model, the images of both archives were compressed using the JPEG 2000 algorithm. TensorFlow DL library was utilized for the proposed model. Due to the minimum codeblock size constraint (see Section II-A), we considered a three-level wavelet decomposition for both image archives ($L = 3$). The codestreams associated with the coarsest wavelet subband ($l = 3$) is used as the input to the proposed approach. The number of transposed convolutional layers (m) is equivalent to the number of wavelet decomposition levels used in the considered image archive. To avoid information loss, we selected the size of the filter as 1×1 with stride 1 and padding 0. The number of filters used for approximating the wavelet subbands is 12×12 and the image is 3×3. For scenario 2, we considered decoding up to $(m-1)$ wavelet decomposition levels. Both image archives were initially divided into three subsets: training (80%), validation (10%) and test (10%) as shown in Table I. Images included in each subset were randomly sampled. The training of the proposed approach was carried out with the stochastic gradient descent (SGD), which uses the Adaptive Moment Estimation (Adam). During training, the Xavier initialization method was used for the parameter initialization. As there is no pretrained models to perform scene classification in the compressed domain, all the experiments were performed starting from scratch. In addition, to achieve accurate performance, experiments were carried out varying learning rate between 0.1 and 0.0001. The performance of the proposed architecture was assessed quantitatively and qualitatively using: 1) classification accuracy; 2) computational time (in sec) of training, validation, and test phases; and 3) root mean square error (RMSE) of the approximated subband images. It is worth noting that computational time of the test phase was considered as classification time. All the experiments were performed with Nvidia Tesla V100.

V. EXPERIMENTAL RESULTS

To evaluate the effectiveness of the proposed approach, we performed several experiments to: 1) assess the quality of the proposed approximated images compared to the decoded wavelet subbands (image); 2) analyze the performance of the proposed approach for scenarios 1 and 2 (mentioned in Sections III-A1 and III-A2); and 3) compare the performance and computational gain with respect to a standard CNN. To this end, we considered two different cases under scenario 1 and 2 where as follows.

1) *Scenario 1:* The coarsest level wavelet subbands are used to approximate the image-level information.
2) *Scenario 1:* The coarsest level wavelet subbands are used to approximate the intermediate finer level wavelet subbands.
3) *Scenario 2:* Decoded finer level wavelet subbands are used to approximate the image-level information.
4) *Scenario 2:* Decoded finer level wavelet subbands are used to approximate intermediate finest level wavelet subbands.

In the first set of experiments, we assess the qualitative as well as quantitative performances of the proposed approach for scene classification on both NWPU-RESISC45 and AID image archives.

A. Qualitative Analysis of the Approximated Images

This section provides a qualitative analysis of the images obtained by the proposed approximation approach for the NWPU-RESISC45 archive. Figs. 6–9 show the approximated images obtained for LL, LH, HL, and HH wavelet subbands for the images of NWPU-RESISC45 archive that are associated with the building class. To qualitatively analyze the efficiency of the proposed approach, we provide the RMSE value between the approximated image and the decoded image. Given a coarser level wavelet subband (64×64) of the image from the NWPU-RESISC45 archive (Scenario 2), one can notice that the proposed approach is efficient to model the finer level wavelet subband (128×128). It converges fast (around Epoch 50) for all the wavelet subbands. The RMSE values obtained for LL subbands are 157.86, 165.62, and 166.96 for Red (R), Green (G), and Blue (B) bands, respectively. Transposed convolution used to approximate the finer level wavelet subbands (image) introduces a loss to the fine-scale detailed information. This is visible from the LL subband finest approximated images (128×128) (see Fig. 6). The RMSE values for the HL (vertical) subbands are 8.38, 8.64, and 8.74 for RGB bands, respectively. In addition, we also notice decreased RMSE values for the detail wavelet subbands (which are LH, HL, and HH) when compared to the approximation wavelet subband (which is LL). Thus, we can see that the transposed convolution efficiently approximates the detail wavelet subbands. The same behavior is replicated when AID image archive is used; however, they are not repeated due to space constraints.

B. Quanitative Results of the Proposed Approximation Approach

This section presents the classification accuracies and the computational time required by the proposed approach. For the following experiments, the feature extraction and classification steps of the proposed approach have been based on the AlexNet model. Table II reports the performance

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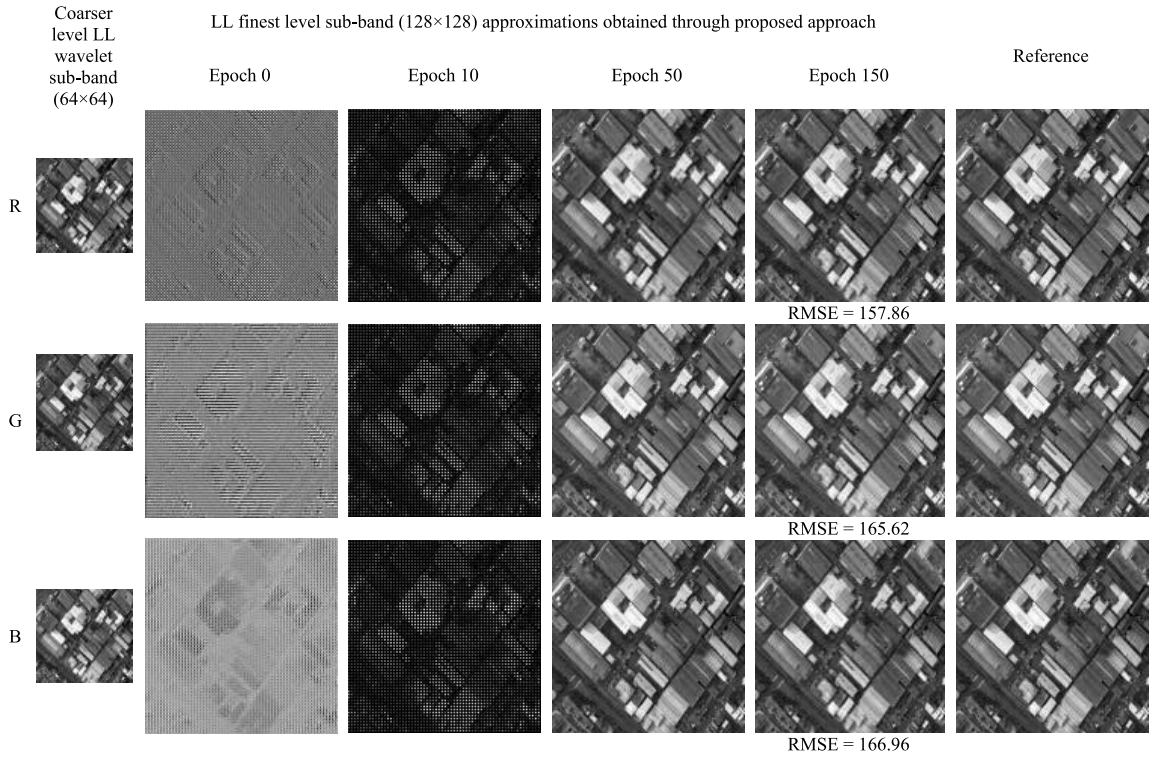


Fig. 6. Qualitative results of subband approximations associated with LL wavelet subband of an image belonging to building category (NWPU-RESISC45 archive).

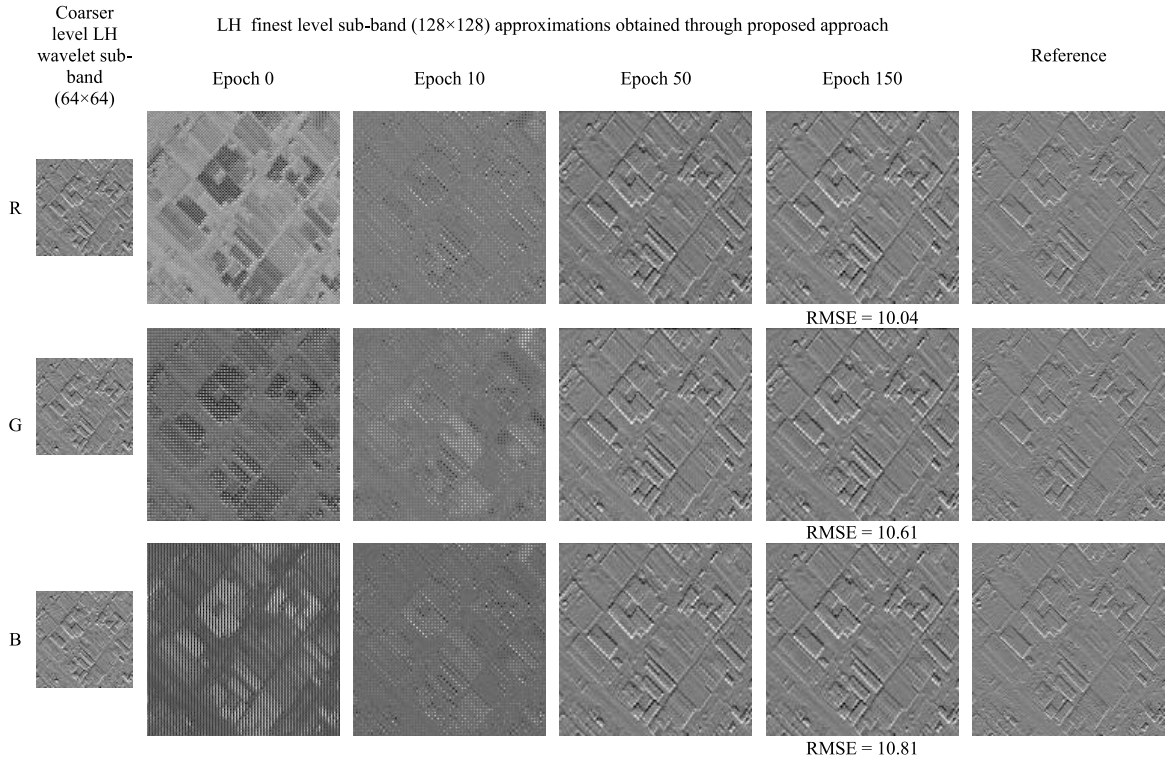


Fig. 7. Qualitative results of subband approximations associated with LH wavelet subband of an image belonging to building category (NWPU-RESISC45 archive).

of the proposed approach (for scenarios 1 and 2) for the NWPU-RESISC45 benchmark archive. It is noted that the computational time includes the decoding time required for the considered images. From the numbers in Table II associated

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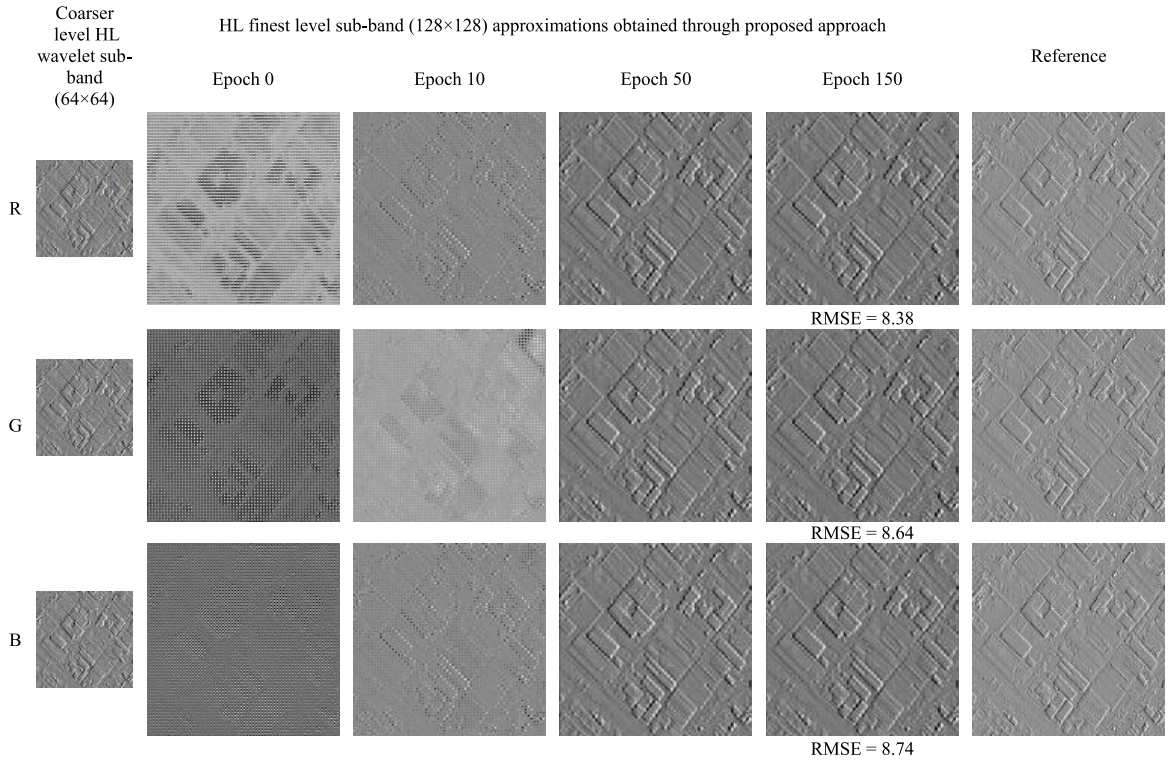


Fig. 8. Qualitative results of subband approximations associated with HL wavelet subband of an image belonging to building category (NWPU-RESISC45 archive).

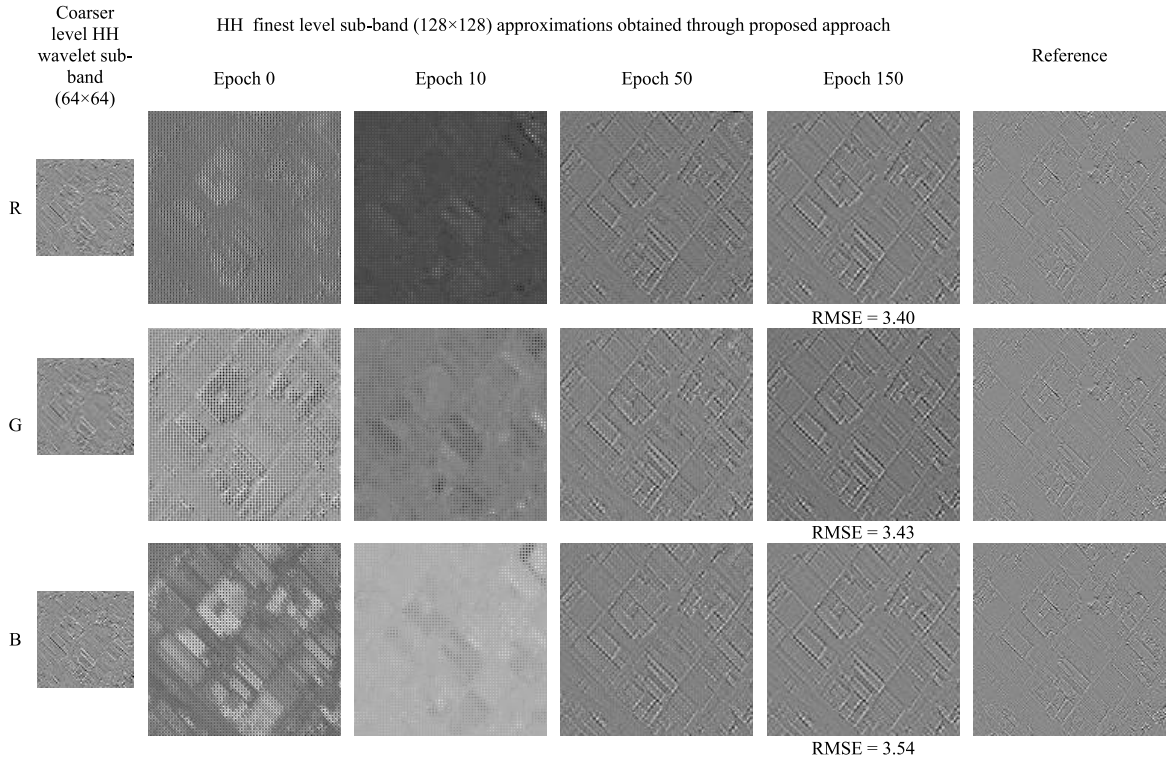


Fig. 9. Qualitative results of subband approximations associated with HH wavelet subband of an image belonging to building category (NWPU-RESISC45 archive).

with Scenario 1, one can notice that the proposed approach employs the coarsest level wavelet subbands (32×32) to approximate: 1) the image (256×256) after applying three transposed convolutional layers; 2) the finest level wavelet

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TABLE II
CLASSIFICATION ACCURACY AND COMPUTATIONAL TIME FOR THE PROPOSED APPROXIMATION APPROACH (NWPU-RESISC45 ARCHIVE)

Proposed Approximation Approach		Accuracy (%)	Computational Time (sec)		
			Train	Validation	Test
Scenario 1	Approximating image $(32 \times 32) \rightarrow (64 \times 64) \rightarrow (128 \times 128) \rightarrow (256 \times 256)$	73.27	8770.76	6.13	5.17
	Approximating finest level wavelet sub-bands $(32 \times 32) \rightarrow (64 \times 64) \rightarrow (128 \times 128)$	74.05	6739.87	5.28	5.68
	Approximating finer level wavelet sub-bands $(32 \times 32) \rightarrow (64 \times 64)$	65.42	568.99	0.38	0.51
Scenario 2	Approximating image $(64 \times 64) \rightarrow (128 \times 128) \rightarrow (256 \times 256)$	80.09	8630.20	106.37	106.51
	Approximating finest level wavelet sub-bands $(64 \times 64) \rightarrow (128 \times 128)$	79.92	8393.79	102.03	101.81
	Approximating image $(128 \times 128) \rightarrow (256 \times 256)$	78.54	8853.99	207.24	206.81

subband (128×128) after applying two transposed convolutional layers; and 3) the finer level wavelet subband (64×64) after applying one transposed convolutional layer. As one can observe, approximating the finest level wavelet subband (128×128) achieves the best classification performance when compared to the other two cases. This is due to the fact that when the coarsest level wavelet subbands are used to approximate image (which requires three transposed convolution layers), the details of the approximated fine-scaled objects are reduced. Nonetheless, when the finest level wavelet subbands (128×128) are approximated (which requires only two transposed convolution layers), we gain in terms of both performance and computational time. Thus, we can conclude that, as the number of layers used for approximation decreases, the performance increases. In the third case, where the finer level subbands (64×64) is approximated (using one transposed convolutional layers), the size of the approximated wavelet subbands does not provide enough image information for accurate classification. Thus, the resulting classification accuracy is the lowest (i.e., 65.42%) when compared to the other two cases.

From the numbers in Table II associated with scenario 2, one can see that the proposed approach has used the decoded finer level wavelet subbands (64×64) to approximate: 1) the image (256×256) after applying two transposed convolutional layers and 2) the finest level wavelet subbands (128×128). In the third case, the proposed approach uses the decoded finest level wavelet subbands (128×128) to approximate the image (256×256). As one can see, all the three cases report almost similar classification accuracies with very small differences. However, if we compare the computational times, we can observe that the training time required for approximating the finest level wavelet subbands is lower than the time required to approximate the images. The training time required when the finest level wavelet subbands are used is 6739.87 s. In the classification phase, the proposed approach takes 206.81 s when the image is approximated after decoding two wavelet decomposition levels. In the first case, where the image is approximated using the finer level wavelet subbands (64×64), the required computational time is only 106.51 s. The overall gain is achieved when the finest level wavelet subbands (128×128) is approximated.

When we compare scenarios 1 and 2 (see Table II), we can notice that the proposed approach attains good classification accuracies when the finest level wavelet subbands are used. If we analyze the performance of the proposed approach when the finest level wavelet subbands (128×128) are obtained, it achieves an accuracy of 74.04% when the coarsest level wavelet subbands are used with a required classification time (i.e., test time) as 5.68 s. In the other case, approximating the finest level wavelet subbands (128×128) after decoding results in 79.92% classification accuracy with a higher computational time of 101.81 s. We can observe that the proposed approach obtains accuracy of 74.05% when only the coarsest level wavelet subbands are used with a significantly reduced computational time 5.68 s. If we perform one-level wavelet decoding to obtain the finer level (64×64) wavelet subbands, which is used to approximate the finest level wavelet subbands (128×128), we notice an increase of 5.87% in classification accuracy. This shows that the proposed approach achieves reasonable classification performance with the coarsest wavelet subbands. Thus, the experimental results demonstrate that the finest level wavelet subbands (partially decoded domain) provide sufficient information for an efficient scene classification with reduced computational time.

Table III reports the performance of the proposed approach on the AID benchmark archive. From Table III (Scenario 1 and 2), the proposed approach employs the coarsest and the decoded finer level wavelet subbands (75×75) to approximate the finer level wavelet subbands (the image itself). While analyzing the part of Table III associated with scenario 1, we can notice that the proposed approach employs the coarsest level wavelet subband to approximate: 1) the image level information (600×600); 2) the finest level wavelet subbands (300×300); and 3) the finer level wavelet subband (150×150). From the results, we can observe an accuracy of 77.34% when we approximate the finest level wavelet subband (150×150). This is because the size of the coarsest level wavelet subband employed provides sufficient information to approximate finer level subbands (150×150) without large information loss with only one transposed convolutional layer. However, when we use two or more transposed convolutional layers to approximate finer level wavelet subbands (the image itself), the accuracy is reduced. This is because the coarsest level

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TABLE III
CLASSIFICATION ACCURACY AND COMPUTATIONAL TIME FOR THE PROPOSED APPROXIMATION APPROACH (AID ARCHIVE)

Proposed Approximation Approach		Accuracy (%)	Computational Time (sec)		
			Train	Validation	Test
Scenario 1	Approximating image $(75 \times 75) \rightarrow (150 \times 150) \rightarrow (300 \times 300) \rightarrow (600 \times 600)$	74.64	14539.87	14.26	14.03
	Approximating finest level wavelet sub-bands $(75 \times 75) \rightarrow (150 \times 150) \rightarrow (300 \times 300)$	76.92	13598.14	13.87	14.91
	Approximating finer level wavelet sub-bands $(75 \times 75) \rightarrow (150 \times 150)$	77.34	10115.91	8.62	9.90
Scenario 2	Approximating image $(150 \times 150) \rightarrow (300 \times 300) \rightarrow (600 \times 600)$	79.91	14183.46	253.98	279.28
	Approximating finest level wavelet sub-bands $(150 \times 150) \rightarrow (300 \times 300)$	79.24	13847.33	224.36	227.34
	Approximating image $(300 \times 300) \rightarrow (600 \times 600)$	78.52	14964.64	326.34	331.65

TABLE IV
CLASSIFICATION ACCURACY AND COMPUTATIONAL TIME FOR THE PROPOSED APPROXIMATION APPROACH AND A STANDARD CNN (NWPU-RESISC45 ARCHIVE)

Model	Method		Accuracy (%)	Computational Time (sec)		
				Train	Validation	Test
AlexNet	Proposed Approximation Approach	Approximating finest level wavelet sub-bands $(32 \times 32) \rightarrow (64 \times 64) \rightarrow (128 \times 128)$	74.05	6739.87	5.28	5.68
		Approximating finest level wavelet sub-bands $(64 \times 64) \rightarrow (128 \times 128)$	79.92	8393.79	102.03	101.81
	Standard CNN	Fully decompressed image (256×256)	80.11	7478.89	305.13	306.24
		Without any decompression (32×32)	54.01	314.20	0.13	0.12
ResNet50	Proposed Approximation Approach	Approximating finer level wavelet sub-bands $(75 \times 75) \rightarrow (150 \times 150)$	85.91	16953.31	15.61	16.01
		Approximating finest level wavelet sub-bands $(150 \times 150) \rightarrow (300 \times 300)$	93.98	18992.71	124.32	125.64
	Standard CNN	Fully decompressed image (600×600)	94.85	17234.51	326.63	325.98
		Without any decompression (75×75)	76.31	763.24	3.64	3.98

wavelet subbands (75×75) introduce checkerboard artifacts when two or more transposed transposed convolutional layers are included. In addition, the training and classification times required are 10115.91 and 9.90 s, respectively. Thus, it requires minimum decoding, which reduced the additional overhead required before classification.

In the part of Table III associated with scenario 2, the proposed approach employs: 1) the finer level wavelet subbands (150×150) to approximate the image level information (600×600); 2) the finer level wavelet subbands (150×150) finest level wavelet subbands (300×300); and 3) the finest level wavelet subbands (300×300) to approximate the image (600×600). If we compare the classification accuracies, we can observe that the highest classification accuracy of 79.91% is achieved when the finer level wavelet subbands (150×150) are used to approximate the image (600×600). However, the training time required to approximate the finest level wavelet (300×300) from the finer level wavelet subbands is lower when compared to the other two cases and the classification accuracy is 79.24% which is very close to the highest one. In addition, this last case has also the lowest classification time. From the experimental results, we can conclude that, if the size of the coarsest level wavelet subbands is large enough (e.g., as in the case of (75×75) AID archive), the proposed approach requires only one approximation level to achieve an acceptable classification accuracy.

C. Comparison of the Proposed Approach With a Standard CNN

In this section, we compare the effectiveness of the proposed approach with: 1) a standard-CNN model where full decompression of images is required and 2) a standard-CNN model that takes as input the coarsest level wavelet subbands (which can be obtained from the codestreams of the compressed image). For the following experiments, the feature extraction and classification parts are based on the ResNet50 model. Tables IV and V report the classification accuracies and computational times for the NWPU-RESISC45 and AID image archives, respectively. It is worth noting that during classification the proposed approach requires only the codestreams associated with the coarsest level wavelet subbands, whereas the standard-CNN model requires the fully decompressed images. By analyzing the tables, one can observe that the computational time required by the proposed approach is significantly reduced when compared to that of the standard-CNN model. In addition, we can also notice that the proposed approach attains almost similar classification accuracies when compared to the standard-CNN model that uses fully decompressed images. On the contrary, if we perform classification using the coarsest level wavelet subbands, the classification accuracy is significantly reduced.

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TABLE V

CLASSIFICATION ACCURACY AND COMPUTATIONAL TIME FOR THE PROPOSED APPROXIMATION APPROACH AND A STANDARD CNN (AID ARCHIVE)

Model	Method		Accuracy (%)	Computational Time (sec)		
				Train	Validation	Test
AlexNet	Proposed Approximation Approach	Approximating finer level wavelet sub-bands $(75 \times 75) \rightarrow (150 \times 150)$	77.34	10115.91	8.62	9.90
		Approximating finest level wavelet sub-bands $(150 \times 150) \rightarrow (300 \times 300)$	79.24	13847.33	224.36	227.34
	Standard CNN	Fully decompressed image (600×600)	79.54	12582.21	412.37	422.84
		Without any decompression (75×75)	61.91	946.23	7.90	8.25
ResNet50	Proposed Approximation Approach	Approximating finer level wavelet sub-bands $(75 \times 75) \rightarrow (150 \times 150)$	84.92	17256.34	14.32	14.13
		Approximating finest level wavelet sub-bands $(150 \times 150) \rightarrow (300 \times 300)$	92.24	b24356.75	298.26	299.50
	Standard CNN	Fully decompressed image (600×600)	93.01	21731.25	443.91	443.12
		Without any decompression (75×75)	69.78	1231.24	13.56	13.14

TABLE VI

LIST OF SYMBOLS USED IN THIS ARTICLE

Symbol	Description
$\mathbf{X}$	an archive that consists of *N* JPEG 2000 compressed images
X_i	i^{th} image in the archive $\mathbf{X}$
$\mathbf{Y}$	set of class labels associated with archive $\mathbf{X}$
y_i	i^{th} class label in set $\mathbf{Y}$
L	number of wavelet decomposition levels used in archive $\mathbf{X}$
G^L	the coarser approximation, horizontal, vertical and diagonal sub-bands of an image X_i at L^{th} wavelet decomposition level
A^{L-1}	the finer approximation, horizontal, vertical and diagonal sub-bands of an image X_i at $L-1^{th}$ wavelet decomposition level
$\mathbf{C}$	sparse matrix
S	stride used in the DNN
P	padding used in the DNN
A_{size}^{L-1}	approximated wavelet sub-band size
A_{size}^{L}	coarser wavelet sub-band size
k	considered kernel
$\mathcal{L}_{total}$	total loss
M	number of rows in the considered wavelet sub-band
N	number of columns in the considered wavelet sub-band
$\mathcal{L}_{approximation}$	approximation loss
$\mathcal{L}_{classification}$	classification loss
$\hat{y}_i$	predicted class label
D^i	decoded wavelet sub-band

By analyzing the AlexNet model results for NWPU-RESISC45 archive (see Table IV), we can notice that the classification accuracy obtained using fully decompressed images with a standard CNN is 80.11%, with a classification time (i.e., test time) of 306.24 s. The proposed approach results in a very similar classification accuracy of 79.92% when only one level of decoding is performed with a lower classification time of 101.81 s. When the coarsest level wavelet subbands (32×32) are used to approximate finest level wavelet subbands (128×128), the required classification time is of more than an order of magnitude smaller at the cost of almost 5% lower classification accuracy. When the coarsest level wavelet-subbands are used in the standard CNN, we obtain the lowest classification accuracy with the lowest classification time. By analyzing the ResNet50 model results for the AID archive (see Table IV), the classification accuracy obtained by fully decompressing the images is 94.85% with a classification time of 325.98 s. The proposed approach results again in a very similar classification accuracy of 93.98% by reducing classification time (i.e., test time) to 125.64 s.

By analyzing the AlexNet model results for NWPU-RESISC45 archive (see Table V), we observe that the proposed approach results in a classification accuracy of 77.34% when the coarsest level wavelet subbands are used, with a classification time of 9.90 s. When we compare the performance of the proposed approach with the standard-CNN, although the classification accuracy is reduced by 2.20%, there is a significant gain in terms of the classification time that is reduced to 9.90 s. Also, it is important to note that the proposed approach reaches a classification accuracy of 79.24% which is similar to that obtained by the standard-CNN approach that requires fully decompressed images. By analyzing ResNet50 model results for the AID archive (see Table V), the classification accuracy obtained by fully decompressing the images is 93.01% with

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a computational time of 444.12 s. The proposed approach results in a similar classification accuracy of 92.24% with a computational time 299.50 s. By analyzing the results, one can conclude that the proposed approach minimizes the computational time considerably when compared to the standard-CNN model. In addition, using a powerful CNN model like ResNet50, the performance is also improved. However, this is achieved at the cost of increasing the computational time.

VI. Discussion and Conclusion

In this article, a novel approach has been presented to perform RS image scene classification in the JPEG 2000 compressed domain using DNNs. The proposed approach minimizes the amount of image decoding by a DNN, which approximates the finer level wavelet subbands from the codestreams associated with the coarser level wavelet subbands. To this end, the proposed approach initially takes the codestreams associated with the coarsest level wavelet subbands to approximate the finer level wavelet subbands with a series of transposed convolutional layers. The aim of the transposed convolutional layers is to approximate the finer level wavelet subbands without requiring to decode the images (to obtain the features for scene classification). This significantly reduces the decoding time required for scene classification, which is the dominant aspect while performing scene classification in compressed RS image archives. Then, the features obtained from the finer level wavelet subbands are obtained through the convolutional layers. Then, the proposed approach performs scene classification based on the obtained features. During training, in addition to the classification loss, the approximation loss is also optimized that is calculated between the approximated wavelet coefficients and the original wavelet coefficients. Thanks to the approximation of finer level wavelet subbands, the time required to decode the images is considerably minimized.

Experimental results in terms of scene classification accuracy and computational gain on two benchmark archives demonstrate the effectiveness of the proposed approach. This is mainly related to the significant reduction of the decoding time associated with the use of a large amount of compressed images. Since there is a tradeoff between the computational gain and the classification accuracy based on the number of transposed convolutional layers, one can always choose the number of layers depending on the requirements in computational time and accuracy. The qualitative images obtained from the approximations show that the proposed approach efficiently operates only with the original coarsest level wavelet coefficients as input source. The results obtained from the experiments demonstrate the ability of the proposed approach:

1) to accurately perform image scene classification in the JPEG 2000 compressed domain;
2) to significantly improve the computational gain by minimizing the required amount of decompression compared to the existing scene classification methods (which operates on uncompressed images).

In view of the growth of RS big data archives, this work introduces a research direction for operating scene classification with DNNs directly on the compressed archives. It is noted that the proposed approach is not limited to JPEG 2000 compressed archives but can be directly applied to any image archive that considers wavelet-based compression approach. In addition, the introduced concept can be adapted to be used in the framework of other compression algorithms by properly modifying the technique used for approximating the compressed domain features. As a future development of this work, we plan to explore scene classification in the context of GANs in the compressed domain. Moreover, we plan to study the development of specific models that can extract features within a deeply compressed domain.

Appendix

A list of notation and symbols used throughout this article is provided in Table VI.

References

[1] X. Zhang, J. Cui, W. Wang, and C. Lin, "A study for texture feature extraction of high-resolution satellite images based on a direction measure and gray level co-occurrence matrix fusion algorithm," *Sensors*, vol. 17, no. 7, p. 1474, Jun. 2017.
[2] F.-A. Georgescu, C. Vaduva, D. Raducanu, and M. Datcu, "Feature extraction for patch-based classification of multispectral Earth observation images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 6, pp. 865–869, Jun. 2016.
[3] A. M. Puetz and R. C. Olsen, "Haralick texture features expanded into the spectral domain," *Proc. SPIE*, vol. 6233, May 2006, Art. no. 623311.
[4] C. Dong, J. Liu, F. Xu, and C. Liu, "Ship detection from optical remote sensing images using multi-scale analysis and Fourier HOG descriptor," *Remote Sens.*, vol. 11, no. 13, p. 1529, Jun. 2019.
[5] I. Tekeste and B. Demir, "Advanced local binary patterns for remote sensing image retrieval," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 6855–6858.
[6] J. Hu, G.-S. Xia, F. Hu, H. Sun, and L. Zhang, "A comparative study of sampling analysis in scene classification of high-resolution remote sensing imagery," in *Proc. Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, vol. 7, no. 11, pp. 2389–2392.
[7] J. Zhang, Z. Cheng, and T. Li, "A bag-of-visual words approach based on optimal segmentation scale for high resolution remote sensing image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 1012–1015.
[8] Q. Zhu, Y. Zhong, B. Zhao, G. Xia, and L. Zhang, "Bag-of-visual-words scene classifier with local and global features for high spatial resolution remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 6, pp. 747–751, Jun. 2016.
[9] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.
[10] X. Chen, S. Xiang, C.-L. Liu, and C.-H. Pan, "Vehicle detection in satellite images by hybrid deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 10, pp. 1797–1801, Oct. 2014.
[11] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Convolutional neural networks for image classification and captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 645–657, 2016.
[12] G. J. Scott, M. R. England, W. A. Starms, R. A. Marcum, and C. H. Davis, "Training deep convolutional neural networks for land–cover classification of high-resolution imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 4, pp. 549–553, Apr. 2017.
[13] I. Sevo and A. Avramovic, "Convolutional neural network based automatic object detection on aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 740–744, May 2016.
[14] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

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[15] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
[16] X. Zhang, J. Zou, K. He, and J. Sun, "Accelerating very deep convolutional networks for classification and detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 10, pp. 1943–1955, Oct. 2016.
[17] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.
[18] Y. Jia *et al.*, "Caffe: Convolutional architecture for fast feature embedding," in *Proc. ACM Int. Conf. Multimedia (MM)*, 2014, pp. 675–678.
[19] G.-S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[20] D. Duarte, F. Nex, N. Kerle, and G. Vosselman, "Multi-resolution feature fusion for image classification of building damages with convolutional neural networks," *Remote Sens.*, vol. 10, no. 10, p. 1636, Oct. 2018.
[21] X. Zheng, Y. Yuan, and X. Lu, "A deep scene representation for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4799–4809, Jul. 2019.
[22] G. Sumbul and B. Demir, "A novel multi-attention driven system for multi-label remote sensing image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 5726–5729.
[23] Y. Guo, J. Ji, X. Lu, H. Huo, T. Fang, and D. Li, "Global-local attention network for aerial scene classification," *IEEE Access*, vol. 7, pp. 67200–67212, 2019.
[24] Y. Yu, X. Li, and F. Liu, "Attention GANs: Unsupervised deep feature learning for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 519–531, Jan. 2020.
[25] J. Li, J. Jia, and D. Xu, "Unsupervised representation learning of image-based plant disease with deep convolutional generative adversarial networks," in *Proc. 37th Chin. Control Conf. (CCC)*, Jul. 2018, pp. 1–16.
[26] S. Roy, E. Sangineto, N. Sebe, and B. Demir, "Semantic-fusion GANs for semi-supervised satellite image classification," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 684–688.
[27] D. Lin, K. Fu, Y. Wang, G. Xu, and X. Sun, "MARTA GANs: Unsupervised representation learning for remote sensing image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2092–2096, Nov. 2017.
[28] Y. Duan, X. Tao, M. Xu, C. Han, and J. Lu, "GAN-NL: Unsupervised representation learning for remote sensing image classification," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2018, pp. 375–379.
[29] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.
[30] S. Zhou, C. Deng, B. Zhao, Y. Xia, Q. Li, and Z. Chen, "Remote sensing image compression: A review," in *Proc. IEEE Int. Conf. Multimedia Big Data*, Apr. 2015, pp. 406–410.
[31] D. Taubman and M. Marcellin, *JPEG2000: Image Compression Fundamentals, Standards and Practice*. New York, NY, USA: Springer, 2002. [Online]. Available: https://www.springer.com/gp/book/9780792375197
[32] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.
[33] A. Descampe, C. De Vleeschouwer, P. Vandergheynst, and B. Macq, "Scalable feature extraction for coarse-to-fine JPEG 2000 image classification," *IEEE Trans. Image Process.*, vol. 20, no. 9, pp. 2636–2649, Sep. 2011.
[34] S. Mallat and W. L. Hwang, "Singularity detection and processing with wavelets," *IEEE Trans. Inf. Theory*, vol. 38, no. 2, pp. 617–643, Mar. 1992.
[35] M. Allili, "Wavelet modeling using finite mixtures of generalized Gaussian distributions: Application to texture discrimination and retrieval," *IEEE Trans. Image Process.*, vol. 21, no. 4, pp. 1452–1464, Apr. 2012.
[36] L. Wang, H. Zhong, R. Ranjan, A. Zomaya, and P. Liu, "Estimating the statistical characteristics of remote sensing big data in the wavelet transform domain," *IEEE Trans. Emerg. Topics Comput.*, vol. 2, no. 3, pp. 324–337, Sep. 2014.
[37] S. K. Choy and C. S. Tong, "Statistical wavelet subband characterization based on generalized gamma density and its application in texture retrieval," *IEEE Trans. Image Process.*, vol. 19, no. 2, pp. 281–289, Feb. 2010.
[38] M. H. Pi, C. S. Tong, S. K. Choy, and H. Zhang, "A fast and effective model for wavelet subband histograms and its application in texture image retrieval," *IEEE Trans. Image Process.*, vol. 15, no. 10, pp. 3078–3088, Oct. 2006.
[39] A. Teynor, W. Müller, and W. Kowarschick, "Compressed domain image retrieval using JPEG2000 and Gaussian mixture models," in *Proc. Int. Conf. Adv. Vis. Inf. Syst.*, 2005, pp. 132–142.
[40] E. de Ves, D. Acevedo, A. Ruedin, and X. Benavent, "A statistical model for magnitudes and angles of wavelet frame coefficients and its application to texture retrieval," *Pattern Recognit.*, vol. 47, no. 9, pp. 2925–2939, Sep. 2014.
[41] D. H. B. Kekre, S. Thepade, P. Mukherjee, M. Kakaiya, S. Wadhwa, and S. Singh, "Image retrieval with shape features extracted using morphological operators with BTC," *Int. J. Comput. Appl.*, vol. 12, no. 3, pp. 1–5, Dec. 2010.
[42] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.
[43] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.



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